

FDE-Associate Foundation Exam

Application-Based MCQ Practice Set

35 Exam-Style Questions | Scenario-Based | Two-Selection | Table-Based

Source basis: What AI Actually Is: A Crash Course — Agent Factory / Panaversity. Questions are original application scenarios derived only from the supplied source.

Section A — Scenario-Based MCQs (1-20)

1. A support team asks a language model to identify the latest outage affecting a vendor. The model gives a polished answer from its existing knowledge. The team needs information from today rather than a general explanation. They want the system to obtain the current vendor status before drafting the response. Which design best matches the course's model of how current information reaches the answer?
 - A. Ask the model to be more confident about its internal knowledge.
 - B. Use a web-search tool, place the returned facts into context, and then let the model generate the response.
 - C. Retrain the model during the conversation with the vendor's latest status.
 - D. Increase the number of tokens in the prompt without adding current evidence.
2. An engineer corrects a model's incorrect product specification in one conversation. The model immediately agrees and uses the corrected value later in that same chat. The engineer then opens a new chat and asks the original question again. The old correction is absent. Which explanation is most consistent with the source?
 - A. The correction permanently changed the model's weights but the new chat selected a different tokenizer.
 - B. The model stored the correction in its weights but forgot the associated prompt.
 - C. The correction was part of the current context; using the model did not retrain its frozen weights.
 - D. The knowledge cutoff moved backward when the new chat began.
3. A company wants its assistant to answer questions about an internal sales dashboard. The model's training data could not have contained yesterday's private figures. The team assumes the model must already know the values but is refusing to reveal them. They need a mechanism that actually gives the model access to those figures. What should they understand first?
 - A. Private data is automatically added to model weights after an employee mentions it.
 - B. The model's frozen weights contain the company's private figures if the figures are recent enough.
 - C. The model can use private figures only when those figures are supplied through the context or an appropriate tool/connector.
 - D. A larger context window automatically creates knowledge that was never supplied.
4. A developer asks an assistant to count letters in a rare invented string. The assistant repeatedly produces the wrong count even though it can generate sophisticated code. The developer concludes that the model is generally incapable of language tasks. The course instead uses this kind of failure to illustrate a more specific limitation. Which diagnosis is most appropriate?
 - A. The model's capabilities are jagged, and exact letter-level operations can be weak because it processes token chunks.
 - B. The model has no training data for any programming task.
 - C. The model always treats each word as one indivisible character.
 - D. The model cannot process invented strings because all tokens must have appeared verbatim in training.

5. A product team is deciding whether a very large context window solves every long-document problem. Their analyst says that once enough tokens fit, unrelated history no longer matters. The team wants to understand why a huge window still has practical limits. Which statement best reflects the course?
- A. A larger window is infinite for practical purposes.
 - B. All content inside a window is equally useful regardless of relevance.
 - C. The window is finite and shared; too much unrelated material can dilute important signal and increase processing cost.
 - D. The context window changes the frozen weights whenever it becomes crowded.
6. A manager notices that a model gives highly confident answers about a niche local business that has almost no online presence. The details are neatly formatted and sound realistic, but the manager cannot verify them. The team needs to interpret the behavior correctly before acting. What is the most source-consistent diagnosis?
- A. Confidence is evidence that the model retrieved the facts from its weights.
 - B. The model may be generating a fluent likely continuation without an internal truth-checker, producing a hallucination.
 - C. The model has probably updated its knowledge cutoff during the answer.
 - D. The model is deliberately hiding a verified answer because the topic is local.
7. A consultant asks an AI, 'Isn't my proposed architecture obviously the right design?' The model immediately agrees and supplies supporting reasons. The consultant wants to reduce the chance that the wording itself steers the response. Which change is most appropriate according to the course's explanation of confidence and agreement?
- A. Use neutral framing and ask the model to evaluate the strongest case on each side.
 - B. Repeat the leading question more forcefully.
 - C. Tell the model that disagreement will be considered a failure.
 - D. Remove all criteria so the model has maximum freedom to agree.
8. An operations analyst observes that a model can write a complex explanation but sometimes fails at a simple three-step logic puzzle. The analyst proposes using success on the complex task as evidence that all simpler tasks are safe. The team needs a better mental model for evaluation. What should they do?
- A. Assume capability is smooth and verify only the complex task.
 - B. Treat the model as jagged: test neighboring and easy-looking tasks instead of inferring broad competence.
 - C. Use confidence of the response as the main quality signal.
 - D. Avoid testing simple tasks because they are less representative of intelligence.
9. A knowledge worker has a long chat containing many unrelated drafts, old decisions, and current requirements. The assistant begins missing important details near the beginning. The worker wants to recover reliability without changing the underlying model. Which intervention follows from the source?
- A. Keep adding more unrelated history so the model has more context.
 - B. Start a fresh chat for the new task and carry forward a concise, relevant summary.
 - C. Assume the model has become tired and wait for it to recover.
 - D. Delete the model's weights and retrain them with the conversation.
10. A team builds a reusable skill containing extensive reference material and instructions. They worry that loading the entire skill into every request will crowd the context window. The system supports loading detailed material only when a request matches the skill. Which design principle is being used?
- A. Knowledge cutoff extension
 - B. Progressive disclosure
 - C. Inference retraining
 - D. Token-free memory

11. A finance team needs an agent to read a spreadsheet, run a calculation, inspect the result, and then decide whether another action is needed. A plain chatbot can describe how to do the calculation but cannot perform it. Which architecture best matches the source's definition of an agent?
- A. A larger prompt with no external actions
 - B. A predictor with tools running a repeated predict-act-observe loop toward a goal
 - C. A model whose weights change after every tool call
 - D. A chatbot that only produces longer explanations
12. An organization connects its agent to an internal database and email system. The security team wants to ensure the agent can reach only the operations it has been explicitly authorized to use. The team also wants a standardized way for connectors to plug into agents. Which statement best fits the source?
- A. A connector grants unrestricted access to the entire application.
 - B. MCP is a shared standard for connectors, while permissions determine what the connector can access.
 - C. MCP is the model's internal truth-checker.
 - D. Connectors work only by changing the model's frozen weights.
13. A model's training corpus ended before a major event. An analyst asks the model for the event's exact outcome without using any current-information tool. The model provides a specific answer. The analyst wants to explain why the answer should not automatically be treated as current knowledge. Which concept is central?
- A. Context window size alone guarantees current facts.
 - B. Knowledge cutoff: information after training ended is not in the frozen weights.
 - C. RLHF guarantees the newest facts are preferred.
 - D. Tokenization automatically imports recent news.
14. An AI assistant receives a request containing a system-level instruction, the user's prompt, a prior transcript, and an attached document. The engineer asks where the model obtains the specifics needed for its next response. Which answer best matches the course?
- A. Only the model's weights; context is ignored.
 - B. Only the last user message.
 - C. The context window containing the available prompt, history, files, tool descriptions, and system instructions.
 - D. Only information retrieved during pretraining.
15. A team is comparing two ways to process a difficult architecture problem. In one approach, the model produces a final answer immediately. In the other, it generates intermediate working before producing the final answer. The team wants to understand what 'reasoning' means mechanically in the source. Which explanation is correct?
- A. Reasoning adds a separate truth-checking organ to the model.
 - B. Reasoning changes the model from prediction into database lookup.
 - C. Reasoning predicts a longer stretch of intermediate working before predicting the final answer; it remains prediction.
 - D. Reasoning permanently modifies the weights for the next task.
16. A team wants to use a reasoning mode for every request, including simple formatting jobs. Their usage budget is limited. The source explains why this can be inefficient. Which decision is most consistent with the material?
- A. Use more reasoning tokens for quick lookups because every task benefits equally.
 - B. Reserve extended reasoning for hard or consequential problems because the extra working consumes time and tokens.
 - C. Avoid reasoning entirely because intermediate working cannot improve answers.
 - D. Use reasoning only when the knowledge cutoff has changed.

17. An evaluator sees a response that is fluent, confident, and highly detailed. No citation or external evidence is available, and the subject is rare. The evaluator is tempted to treat fluency as proof. Which review principle should govern the decision?
- A. Fluency itself is a truth signal.
 - B. Confidence and correctness are mechanically linked.
 - C. The model lacks a separate truth-checker, so the evaluator remains responsible for checking the claim.
 - D. Rare topics are always accurate because the model fills gaps carefully.
18. A developer notices that a long conversation becomes slower and more expensive with each turn. They initially suspect that the model is retaining more information internally. The course gives a different explanation. Which one?
- A. The frozen weights expand with each message.
 - B. The growing transcript is replayed into the context on each turn, increasing the amount of text processed.
 - C. The tokenizer stops working after the first ten turns.
 - D. The model retrains itself after each response.
19. A team wants an assistant to retrieve a document from a company drive and use its contents in a response. The document is not part of the model's training data. Which sequence best represents the source's mechanism?
- A. The model changes its weights to absorb the document, then answers.
 - B. A connector/tool retrieves the document; the result enters context; the model predicts a response from that context.
 - C. The model guesses the document from its name without needing access.
 - D. The document becomes part of pretraining immediately after retrieval.
20. An evaluation program is designed around a difficult task the model performs very well. The team therefore stops testing simpler neighboring tasks. Later, an easy-looking subtask fails unexpectedly. Which lesson from the source most directly explains this failure?
- A. Model capability follows human notions of difficulty.
 - B. A successful hard task proves nearby easy tasks will also succeed.
 - C. Capability is jagged, so verification must cross the boundary into tasks that may look easy.
 - D. Only difficult tasks can expose model errors.

Section A — Answer Key

1. B 2. C 3. C 4. A 5. C 6. B 7. A 8. B 9. B 10. B
11. B 12. B 13. B 14. C 15. C 16. B 17. C 18. B 19. B 20. C

Section B — Two-Selection MCQs (21-28)

21. A product team needs an assistant to work with information that was not in the model's training data. They want the assistant to use current or private material without retraining the underlying model. Which TWO approaches directly fit the source's mechanism? Select TWO options.
- A. Use a web-search tool to fetch current information into context.
 - B. Assume the model's frozen weights already contain the private material.
 - C. Use an authorized connector to retrieve the relevant application data into context.
 - D. Increase the model's confidence setting without adding information.
22. An engineering team is diagnosing why a model sounds certain about a rare technical claim. They want to explain the behavior using the course's mental model rather than assuming the model has verified the claim. Which TWO statements are supported? Select TWO options.
- A. The model has no separate internal truth-checker.
 - B. A confident style can be learned independently of whether the content is true.
 - C. Confidence guarantees that the training corpus contained the exact fact.
 - D. A fluent answer proves the model performed a database lookup.

23. A team is redesigning a long-running assistant workflow because important details are being diluted. They cannot change the model weights. Which TWO actions align with the source? Select TWO options.
- A. Start a fresh chat for a fresh task when the existing context is overloaded.
 - B. Use concise summaries to carry forward only relevant information.
 - C. Keep every unrelated message forever because more context is always better.
 - D. Treat the context window as unlimited storage.
24. A developer wants to explain what makes an agent different from a plain text predictor. Which TWO elements are essential to the source's description? Select TWO options.
- A. The model can call defined tools that perform real actions.
 - B. The system repeats a predict-act-observe loop toward a goal.
 - C. The model must rewrite its weights after every action.
 - D. The model must have a separate human-like truth-checking organ.
25. A team is building a model-selection test. They want to respect the source's description of uneven capability. Which TWO practices are appropriate? Select TWO options.
- A. Test easy-looking neighboring tasks rather than assuming success transfers.
 - B. Re-test the frontier over time because newer models can change capability.
 - C. Use one difficult benchmark as proof of broad competence.
 - D. Treat model confidence as a substitute for evaluation.
26. A developer is estimating token-related constraints for an application. Which TWO statements are directly supported by the source? Select TWO options.
- A. Tokens are the chunks the model reads and writes.
 - B. Token counts affect context capacity and can also be used for cost measurement.
 - C. Every token is exactly one English word.
 - D. Images and audio are outside the token mechanism.
27. A company wants its model to behave more like an assistant than a raw next-text predictor. Which TWO training concepts from the source are relevant? Select TWO options.
- A. Instruction tuning teaches patterns for responding to instructions.
 - B. Human-feedback-based tuning shapes preferred manner and style.
 - C. Inference permanently adds new training examples to the weights.
 - D. A context window replaces all training stages.
28. A team is deciding how to use extended reasoning within a limited budget. Which TWO statements are consistent with the course? Select TWO options.
- A. Reasoning generates additional intermediate working before the final answer.
 - B. Reasoning can improve difficult tasks but still does not create a truth-checker.
 - C. Reasoning makes the model's weights learn from the current problem permanently.
 - D. Reasoning eliminates the need for human verification.

Section B — Answer Key

21. A & C 22. A & B 23. A & B 24. A & B
 25. A & B 26. A & B 27. A & B 28. A & B

Section C — Detailed Table-Based MCQs (29-35)

29. Guidance/Scenario/real life implementation/issue caused: A customer-support system must answer questions about a product whose documentation changes frequently. The base model may have stopped training before the latest revision. The engineering team has two candidate designs and must choose the one that actually supplies fresh information before generation.

Factor	Current Situation	Evidence / Constraint	Implication
Knowledge source	Frozen model only	Training cutoff may predate revision	Latest facts may be absent
Freshness mechanism	Web search	Fetches current pages into context	Current evidence becomes available
Generation	Model response	Uses whatever is in context	Still predicts the answer from supplied text

Question: Which design best addresses the freshness requirement?

- A. Use the frozen model alone and rely on confident wording.
- B. Use web search to fetch current material, then generate from the returned context.
- C. Retrain the model for every support request.
- D. Increase context size without adding current material.

30. Guidance/Scenario/real life implementation/issue caused: An analyst is reviewing a long conversation that contains hundreds of messages. The current task depends on three recent decisions, but the assistant has begun omitting earlier relevant details. The analyst maps the system behavior before deciding whether to restart the conversation.

Factor	Current Situation	Evidence / Constraint	Implication
Chat length	Very large	Transcript grows each turn	More text must be processed
Relevant signal	Small subset	Important details compete with unrelated history	Signal can be diluted
Restart strategy	Fresh chat + summary	Carry only relevant decisions	Smaller, focused context

Question: Which intervention most directly follows from the table and source?

- A. Add more unrelated history.
- B. Start fresh and provide a concise relevant summary.
- C. Retrain the model inside the chat.
- D. Assume the model's weights are degrading.

31. Guidance/Scenario/real life implementation/issue caused: A team is assessing an assistant that performs exceptionally well on common coding tasks but sometimes fails at simple letter counting and rare logic puzzles. The team wants an evaluation policy that reflects uneven capability rather than a single global intelligence score.

Factor	Current Situation	Evidence / Constraint	Implication
Observed task	Common coding	Strong performance	Do not generalize automatically
Observed task	Letter counting	Weak performance	Token-level limitation can matter
Evaluation risk	Single benchmark	Misses neighboring failures	Capability may be jagged
Mitigation	Multiple task types	Different frontier regions tested	Better coverage

Question: Which evaluation approach best matches the source?

- A. Test only the hardest coding task.
- B. Infer all capabilities from one success.
- C. Test across varied and easy-looking tasks, including neighboring failure modes.
- D. Use confidence tone as the evaluation metric.

32. Guidance/Scenario/real life implementation/issue caused: A finance agent must retrieve a spreadsheet, calculate a value, inspect the result, and possibly retrieve another file. The system designer needs to distinguish a plain response generator from an agentic loop and must keep the model's role clear.

Factor	Current Situation	Evidence / Constraint	Implication
Model role	Predict next text/action	Chooses a likely continuation	Can select a tool call
Tool	File read / code execution	Performs external action	Real result returns
Observation	Tool result in context	Becomes available to model	Supports next prediction
Loop	Predict → act → observe	Repeated toward goal	Agent behavior

Question: Which architecture is represented by the table?

- A. A static chatbot with no external action.
- B. A predictor with tools in a repeated predict-act-observe loop.
- C. A model retrained after every tool call.
- D. A database lookup system with no prediction.

33. Guidance/Scenario/real life implementation/issue caused: A platform team is configuring reusable instructions and reference material for specialized tasks. They want detailed knowledge available when needed without placing every file into every request. The design must respect the finite context window.

Factor	Current Situation	Evidence / Constraint	Implication
Knowledge storage	External skill files	Detailed material stays outside active context	Available when relevant
Context entry	Short skill description	Signals when a skill matches	Controls what gets loaded
Loading policy	On demand	Full skill enters context when matched	Reduces unnecessary crowding
Principle	Progressive disclosure	Load only what this moment needs	Preserves context capacity

Question: Which mechanism is being applied?

- A. Progressive disclosure.
- B. Knowledge cutoff extension.
- C. Weight retraining.
- D. Permanent model memory.

34. Guidance/Scenario/real life implementation/issue caused: A compliance reviewer receives a highly polished answer about a rare topic. No tool result or source evidence is present. The reviewer must decide whether the style of the response is enough to establish truth before approving it.

Factor	Current Situation	Evidence / Constraint	Implication
Answer style	Highly confident	Confidence is learned style	Not proof of correctness
Topic	Rare	Training coverage may be thin	Prediction may be weak
Truth checking	No separate checker	Model generates continuation	Human review remains necessary
Risk	Unverified claim	Fluent invention possible	Do not treat tone as evidence

Question: Which approval rule follows most directly?

- A. Approve because confidence indicates strong internal certainty.
- B. Treat fluency as evidence of database retrieval.
- C. Require verification because confidence is not a truth signal.
- D. Approve only if the answer is long enough.

35. Guidance/Scenario/real life implementation/issue caused: An organization connects an agent to an internal tracker and email service. The architecture must let the agent access only the operations granted by the organization while using a standard connector interface. The team documents how data moves after a connector call.

Factor	Current Situation	Evidence / Constraint	Implication
Connector	Tool wired to real app	Permissions are granted by organization	Access is bounded
Standard	MCP	Shared connector plug shape	Common interface
Returned data	Connector result	Placed into context	Model can use it for next prediction
Weights	Frozen	Not changed by connector use	No per-call retraining

Question: Which statement best describes the resulting architecture?

- A. The connector changes the model's weights.
- B. MCP is a truth-checker inside the model.
- C. The connector supplies authorized data through a standard interface, and the result becomes context for the model.
- D. The agent receives unrestricted access to the entire connected application.

Section C — Answer Key

29. B 30. B 31. C 32. B 33. A 34. C 35. C

Source reviewed: Agent Factory / Panaversity, "What AI Actually Is: A Crash Course," including the nine core ideas and the source's practical examples and explanations. The optional product appendix was not used to introduce unrelated product-specific concepts.